① a) $Z = \sum_k \sum_{n_k} e^{-\varepsilon(k) n_k / k_B T} = \sum_k \sum_{n_k} \left(e^{-\varepsilon(k)/k_B T}\right)^{n_k}$

$\sum_n x^n = \frac{1}{1-x}$ so $Z = \sum_k \frac{1}{1-e^{-\varepsilon(k)/k_B T}}$

Transform to integral

$$Z = \left(\frac{L}{2\pi}\right)^3 \int_{-\infty}^{\infty} \frac{1}{1-e^{-\varepsilon(k)/k_B T}} d^3k$$ switch to spherical coordinates

$$Z = \left(\frac{L}{2\pi}\right)^3 4\pi \int_0^{\infty} dk\, k^2 \frac{1}{1-e^{-\varepsilon(k)/k_B T}}$$ $L^3 = V$ (volume)

Then $F = -k_B T \ln Z$

$$F = -k_B T \frac{4\pi}{(2\pi)^3} V \int_0^{\infty} dk\, k^2 \frac{1}{1-e^{-\varepsilon(k)/k_B T}}$$

$$= +\frac{k_B T V}{2\pi^2} \int_0^{\infty} k^2 \ln\left(1-e^{-\varepsilon(k)/k_B T}\right) dk$$

b) $$F = \frac{k_B T V}{2\pi^2} \int_0^{k_{max}} k^2 \ln\left(1-e^{-\hbar s k/k_B T}\right) dk$$ (phonons)

$$+ \frac{k_B T V}{2\pi^2} \int_{k_{max}}^{\infty} k^2 \ln\left(1-e^{-\left(\Delta + \frac{\hbar^2 (k-k_0)^2}{2\mu}\right)/k_B T}\right) dk$$ (rotons)

F_{phonon}: let $x = \frac{\hbar s k}{k_B T}$ then

$$F_{phonon} = \left(\frac{k_B T}{\hbar s}\right)^3 \frac{k_B T V}{2\pi^2} \int_0^{x_{max}} dx\, x^2 \ln\left(1-e^{-x}\right)$$

where $x_{max} = \frac{\hbar s}{k_B T} k_{max}$ low T means $\frac{\hbar s}{k_B T} k_{max} \gg 1$

So we can replace x_{max} with ∞ in the integral. The integral then evaluates to $\frac{-\pi^4}{45}$ and $F_{phonon} = \frac{-k_B{}^4 T^4 V}{90 \hbar^3 s^3}$

Now for F_{roton}:

$$F_{roton} = \frac{k_B T V}{2\pi^2} \int_{k_{max}}^{\infty} k^2 \ln\left(1 - e^{-\Delta/k_B T} e^{-\frac{\hbar^2 (k-k_0)^2}{2\mu k_B T}}\right) dk$$

let $u = k - k_0$, $k = u + k_0$, $du = dk$

$$F_{roton} = \frac{k_B T V}{2\pi^2} \int_{k_{max}-k_0}^{\infty} (u+k_0)^2 \ln\left(1 - e^{-\Delta/k_B T} e^{-\frac{\hbar^2 u^2}{2\mu k_B T}}\right) du$$

low T means $\frac{\hbar^2 u^2}{2\mu k_B T} >> 1$

Then we can expand ln $(\ln(1+x) \sim x)$

and $k_{max} - k_0 \to -\infty$ $(k_{max} < k_0)$

$$F_{roton} = -\frac{k_B T V}{2\pi^2} \int_0^{\infty} (u+k_0)^2 e^{-\Delta/k_B T} e^{\frac{-\hbar^2 u^2}{2\mu k_B T}} du$$

$$F_{roton} = -\frac{k_B T V}{2\pi^2} e^{-\Delta/k_B T} \int_{-\infty}^{\infty} \Big(\underbrace{u^2 e^{\frac{-\hbar^2 u^2}{2\mu k_B T}}}_{(1)} + \underbrace{k_0^2 e^{\frac{-\hbar^2 u^2}{2\mu k_B T}}}_{(2)} + \underbrace{2k_0 u e^{\frac{-\hbar^2 u^2}{2\mu k_B T}}}_{(3)} \Big) du$$

(1) $\frac{\sqrt{\pi}}{4\left(\frac{\hbar^2}{2\mu k_B T}\right)^{3/2}}$ (2) $\frac{k_0^2 \sqrt{\pi}}{\left(\frac{\hbar^2}{2\mu k_B T}\right)^{1/2}}$ (3) $\frac{k_0}{\left(\frac{\hbar^2}{2\mu k_B T}\right)}$

since $\frac{\hbar^2}{2\mu k_B T}$ is large, only keep (2)

then $$F_{roton} = \frac{-k_B T V k_0^2}{4\pi^2} e^{-\Delta/k_B T} \left(\frac{2\pi\mu k_B T}{\hbar^2}\right)^{1/2}$$

(2) a) Because they are harmonic oscillators

$$\mathcal{E}(n_1) = (n_1 + \tfrac{1}{2})\hbar\omega \qquad \mathcal{E}(n_2) = (n_2 + \tfrac{1}{2})\hbar\omega$$

and $E = \mathcal{E}(n_1) + \mathcal{E}(n_2) = (n_1 + n_2 + 1)\hbar\omega$

b) $Z_1 = \sum_{\{s\}} e^{-E_s/k_BT}$ for each particle so $Z_T = \frac{1}{2} Z_1^2 = Z$

$$Z_1 = \int \frac{dp\,dx}{2\pi\hbar} e^{\frac{-p^2}{2mk_BT}} e^{-\frac{m\omega^2x^2}{2k_BT}}$$

$$= \frac{1}{2\pi\hbar} \underbrace{\int_{-\infty}^{\infty} dp\, e^{\frac{-p^2}{2mk_BT}}}_{\sqrt{2\pi m k_B T}} \underbrace{\int_{-\infty}^{\infty} dx\, e^{-\frac{m\omega^2x^2}{2k_BT}}}_{\sqrt{\frac{2\pi k T}{m\omega^2}}}$$

$$Z_1 = \frac{1}{2\pi\hbar} \frac{2\pi m k_B T}{m\omega} = \frac{k_BT}{\hbar\omega} \qquad \text{and} \qquad Z = \frac{k_B^2T^2}{2\hbar^2\omega^2}$$

c) $$Z_B = e^{-\hbar\omega}\left\{\sum_{n_1<n_2} e^{-\beta\hbar\omega(n_1+n_2)} + \sum_{n_1=n_2} e^{-\beta\hbar\omega(n_1+n_2)}\right\}$$

must add in the diagonal terms since bosons can be in same state

Sum over all n_1, n_2 and divide by 2

$$Z_B = \frac{e^{-\hbar\omega\beta}}{2}\left\{\sum_{n_1} e^{-\beta\hbar\omega n_1} \sum_{n_2} e^{-\beta\hbar\omega n_2} + \sum_n e^{-2\beta\hbar\omega n}\right\}$$

$$\sum_n e^{-\beta\hbar\omega n} = \frac{1}{1-e^{-\beta\hbar\omega}} \qquad \text{so}$$

$$Z_B = \frac{e^{-\hbar\omega\beta}}{2}\left\{\left(\frac{1}{1-e^{-\beta\hbar\omega}}\right)^2 + \frac{1}{1-e^{-2\beta\hbar\omega}}\right\}$$

② c) cont...

$$Z_B = \frac{e^{-\hbar\omega\beta}}{2}\left\{ \frac{1-e^{-2\beta\hbar\omega} + (1-e^{-\beta\hbar\omega})^2}{(1-e^{-\beta\hbar\omega})^2(1-e^{-2\beta\hbar\omega})} \right\}$$

$$= \frac{e^{-\hbar\omega\beta}}{2}\left\{ \frac{1-e^{-2\beta\hbar\omega} + 1 - 2e^{-\beta\hbar\omega} + e^{-2\beta\hbar\omega}}{(1-e^{-\beta\hbar\omega})^2(1-e^{-2\beta\hbar\omega})} \right\}$$

$$= \frac{e^{-\hbar\omega\beta}}{2}\left\{ \frac{2-2e^{-\beta\hbar\omega}}{(1-e^{-\beta\hbar\omega})^2(1-e^{-2\beta\hbar\omega})} \right\}$$

$$= \frac{e^{-\beta\hbar\omega}}{(1-e^{-\beta\hbar\omega})(1-e^{-2\beta\hbar\omega})} = Z_B$$

d) This is the same as Z_B but subtract the diagonal terms

$$Z_F = \frac{e^{-\hbar\omega\beta}}{2}\left\{ \frac{1-e^{-2\beta\hbar\omega} - 1 + 2e^{-\beta\hbar\omega} - e^{-2\beta\hbar\omega}}{(1-e^{-\beta\hbar\omega})^2(1-e^{-2\beta\hbar\omega})} \right\}$$

$$Z_F = \frac{e^{-\hbar\omega\beta}}{2}\left\{ \frac{2e^{-\beta\hbar\omega} - 2e^{-2\beta\hbar\omega}}{(1-e^{-\beta\hbar\omega})^2(1-e^{-2\beta\hbar\omega})} \right\}$$

$$Z_F = \frac{e^{-2\hbar\omega\beta}}{(1-e^{-\beta\hbar\omega})(1-e^{-2\beta\hbar\omega})}$$